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RESEARCH ARTICLE

Multimodal Autism Spectrum Disorder Method Using GCN With Dual Transformers

TIANMING SONG[1], ZHE REN[1], JIAN ZHANG[1], YAWEI QU[2,3], YINGYING CUI[3], AND ZHENGDA LIANG[4]

[1]School of Integrated Circuit, Wuxi Vocational College of Science and Technology, Wuxi 214028, China
[2]Department of Control Science and Engineering, Harbin Institute of Technology, Harbin, Heilongjiang 150001, China
[3]Gastroenterology Department, Beijing Mentougou District Hospital, Beijing 102300, China
[4]School of Clinical Medicine, Qilu Medical University, Zibo, Shandong 255300, China

Corresponding author: Zhengda Liang (shancao44611760621@163.com)

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ABSTRACT Autism spectrum disorder (ASD) presents diagnostic challenges due to its heterogeneous nature, necessitating advanced methods for accurate identification. This study proposes a novel diagnostic approach that integrates Graph convolutional networks (GCN) with dual transformer architectures, optimized through a Co-training strategy. The first Transformer is dedicated to extracting intricate temporal features from fMRI data, which are crucial for understanding brain activity over time. The second Transformer is employed to enhance the fusion of these temporal features with spatial features learned by the GCN, effectively combining both dimensions of the neuroimaging data. Co-training is introduced to simultaneously harness both functional magnetic resonance imaging and structural magnetic resonance imaging data, improving the model's capacity to generalize across different datasets. This comprehensive method was rigorously evaluated using the ABIDE-I and ABIDE-II datasets, with performance assessed through nested ten-fold cross-validation and leave-one-out cross-validation. The experimental results reveal that our approach significantly outperforms existing baseline and state-of-the-art models. The method achieves 79.47% of accuracy, 78.97% precision, 82.11% recall, and 0.85 of AUC metrics. These findings highlight the robustness of combining dual transformer models with GCNs and Co-training, providing a powerful framework for ASD classification.

INDEX TERMS Autism spectrum disorder, graph convolutional network, multimodal, medical artificial intelligence, transformer.

I. INTRODUCTION

Autism spectrum disorder (ASD) is a complex neurodevelopmental condition characterized by persistent challenges in social interaction, communication, and behavior [1]. The prevalence of ASD has been rising globally, making it a significant public health concern. Early and accurate diagnosis of ASD is crucial for providing timely interventions, which can significantly improve the quality of life for individuals with ASD. However, the heterogeneous nature of ASD, with its wide range of symptoms and severity, poses considerable challenges for diagnosis, often leading to delays and misdiagnoses.

In recent years, machine learning (ML) [2] and deep learning (DL) [3] techniques have emerged as powerful tools for enhancing the diagnosis of ASD. These methods have been applied to neuroimaging data, particularly functional MRI (fMRI) [4] and structural MRI (sMRI) [5], to identify patterns that distinguish individuals with ASD from neurotypical controls. Machine learning models, such as support vector machines (SVM) [6] and random forests, have shown promise in classifying ASD based on extracted

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features. Meanwhile, deep learning approaches, including convolutional neural networks (CNN) [7] and graph convolutional networks (GCN), have been increasingly utilized to automatically learn and extract complex patterns from high-dimensional neuroimaging data, leading to improved diagnostic accuracy.

Despite the advancements, several challenges remain in the application of ML and DL to ASD diagnosis. The variability in neuroimaging data across different sites and populations can affect the generalizability of models. Moreover, many existing methods either focus solely on spatial or temporal features, neglecting the intricate interplay between these dimensions. To address these challenges, our study proposes a novel approach that integrates GCN and Transformer architectures with a co-training strategy to capture both local and global dependencies in the data. This method enhances the robustness and accuracy of ASD classification by leveraging multimodal neuroimaging data, offering a more comprehensive understanding of the underlying neural mechanisms. The main contributions of this paper are as follows:

- **Transformer for fMRI feature extraction:** We introduce the use of a Transformer model specifically for extracting complex temporal features from fMRI data, allowing for more detailed and accurate characterization of brain activity.
- **Co-training strategy for multimodal data:** We propose a co-training approach that allows the model to simultaneously learn from functional and structural MRI data, improving the model's ability to generalize across different datasets.
- **Transformer for GCN fusion:** We implement a second Transformer model to enhance the fusion of features learned by the GCN, effectively integrating spatial and temporal information for improved classification performance.
- **Superior experimental results:** Through extensive experiments on the ABIDE-I and ABIDE-II datasets, our method consistently outperforms several baseline and state-of-the-art approaches, demonstrating its robustness and effectiveness in diagnosing ASD.

The remainder of this paper is structured as follows: Section II provides a detailed review of related work in ASD diagnosis using machine learning and deep learning approaches. Section III describes the datasets, preprocessing steps, and the proposed methodology, including the model architecture and training process. Section IV presents the experimental settings and results, followed by a discussion in Section V that provides both quantitative and qualitative analysis of the experimental results, highlighting the strengths and implications of our proposed method. Finally, Section VI concludes the paper by summarizing the key findings, discussing the limitations of the study, and suggesting potential directions for future research.

II. RELATED WORK

Several studies have focused on utilizing a single modality, typically fMRI or sMRI, for the diagnosis of ASD. Wang et al. [8] introduced a method called Multi-site Clustering and Nested Feature Extraction to address the variability in multi-site fMRI data. Their approach clusters subjects and extracts key features using a nested singular value decomposition strategy. This method enhances the robustness of ASD detection, outperforming several existing techniques on the ABIDE dataset, particularly by identifying discriminative functional connectivities in critical brain networks like the default mode and salience networks. Feng and Xu [9] proposed an advanced CNN specifically designed to analyze resting-state functional MRI data for early detection of ASD in children. Their study utilized a comprehensive preprocessing pipeline to convert 4D fMRI data into 2D echo planar images, which were then used to train the CNN model. Ma et al. [10] introduced a method that integrates multiple functional connectivity data from resting-state fMRI with GCN for ASD diagnosis. By combining full-brain and key subnetwork connectivity data, their approach significantly improved classification accuracy, highlighting the importance of capturing detailed brain connectivity patterns for effective ASD diagnosis. Mishra and Pati [11] proposed a framework that utilizes sMRI data to detect ASD through an ensemble of deep convolutional neural networks. This approach uniquely combines the same DCNN model with different optimizers and employs on-the-fly data augmentation to prevent overfitting. The proposed method demonstrated superior performance, achieving up to 81.35% accuracy, significantly outperforming previous sMRI-based methods on large, heterogeneous datasets.

Integrating multiple data modalities has become increasingly popular, as it provides a more comprehensive view of the brain's structure and function, thereby improving diagnostic accuracy. Han et al. [12] proposed a framework that combines electroencephalogram and eye-tracking data using stacked denoising autoencoders for ASD diagnosis. Their method effectively captures the complementary information between neurophysiological and behavioral data, improving the identification of ASD in children. The study demonstrated superior performance compared to single-modality methods, highlighting the advantages of multimodal fusion in ASD diagnosis. Wang et al. [13] developed a multimodal framework that integrates functional MRI data with non-imaging demographic information using a weight-learning graph convolutional network. Their approach constructs a population graph to represent subjects, where the edges are weighted based on non-imaging data like age and gender. By incorporating residual connections and an EdgeDrop strategy, the model effectively reduces overfitting and oversmoothing, achieving substantial performance improvements in ASD diagnosis across different sites. Lakhan et al. [14] proposed a federated learning framework combined with CNN-LSTM to diagnose ASD using multimodal datasets. This approach

ensures data privacy across different labs while achieving high detection accuracy, demonstrating the potential of distributed learning in ASD diagnosis.

Recent advancements in transformer models have shown significant promise in the diagnosis of ASD, particularly in handling the complex temporal and spatial relationships within neuroimaging data. Wang et al. [15] introduced a Residual Graph Transformer Network to enhance the prediction of ASD using functional connectivity (FC) data derived from resting-state fMRI. Their model incorporates a Graph Encoder to capture long-range dependencies and residual learning to address the vanishing gradient problem in deep networks. Alharthi and Alzahrani [5] proposed a framework that combines 3D-CNN with vision transformers for ASD diagnosis. Using multi-slice MRI data, their approach achieved state-of-the-art results, highlighting the effectiveness of combining CNNs with transformers in neuroimaging.

In summary, both single-modality and multi-modality approaches have significantly advanced ASD diagnosis, with deep learning methods like GCNs and Transformers showing strong potential in neuroimaging analysis. However, challenges remain in integrating multiple data types and effectively modeling temporal dynamics. Our study addresses these issues by combining GCN and Transformer architectures with a co-training strategy, demonstrating superior performance and improved generalizability in ASD classification using fMRI and sMRI data. This approach enhances both accuracy and robustness.

III. MATERIALS AND METHODS

In this study, we propose a novel multimodal approach for the diagnosis of ASD that integrates GCNs with dual Transformer architectures, optimized through a Co-training strategy. The model is designed to leverage both fMRI and sMRI data, capturing the complex spatial and temporal dynamics within the brain. The overall architecture begins with the preprocessing of fMRI and sMRI data. For fMRI, temporal features are extracted using a Transformer model, while spatial features from sMRI are extracted using a CNN. These features are then fused and input into a GCN, which is further enhanced by a second Transformer-based self-attention mechanism. This combination allows the model to capture both local and global dependencies in the neuroimaging data. Figure 1 shows the overall flow of the proposed method in this paper.

A. DATASETS AND PREPROCESSING

The current research utilizes the Autism Brain Imaging Data Exchange (ABIDE) dataset [16], [17], acclaimed for its extensive compilation of neuroimaging data. which includes ABIDE-I and ABIDE-II. This dataset incorporates fMRI and sMRI scans from a diverse array of participants. Specifically, ABIDE-I includes data from 419 individuals diagnosed with autism spectrum disorder and 530 neurotypical controls. ABIDE-II contains 92 ASD and 103 neurotypical controls. This substantial participant pool, sourced from multiple international research facilities, enhances the statistical robustness of our analysis and supports broader generalizability of the findings. The ABIDE datasets are provided to researchers to encourage data sharing and collaborative research, ultimately advancing the understanding of autism. This study adheres to all applicable ethical standards. We utilized the publicly available ABIDE dataset, for which ethical approval was not necessary, as confirmed by the license associated with the open-access data, given that each site's local IRB had previously approved it. All data were collected with informed consent from the participants. For imaging analysis, the Harvard - Oxford (HO) [18] brain atlas is employed to define regions of interest (ROI) [19] across the brain, ensuring consistent and reliable measurement of neural structures and functions across subjects.

The preprocessing [20] pipeline is meticulously designed to maximize data integrity and comparability across scans. fMRI data undergo several stages of preprocessing: initial slice timing correction adjusts for temporal differences in image acquisition, followed by rigorous head motion correction to minimize artifacts from participant movements. Spatial normalization aligns each individual's fMRI data to a standardized Montreal Neurological Institute (MIN) space [21], enabling effective group-level comparisons. Subsequent image smoothing, using a Gaussian kernel with a 6mm full width at half maximum, enhances the signal-to-noise ratio, crucial for detecting subtle neural activities. Nuisance regression concludes the fMRI preprocessing, removing unwanted variability from motion and other physiological sources to refine data quality.

The sMRI data preprocessing involves skull stripping to focus analyses exclusively on brain tissues, followed by detailed segmentation that categorizes brain tissue into gray matter, white matter, and cerebrospinal fluid. This segmentation provides essential structural details needed for further morphometric analysis. Normalization to MNI space ensures consistent anatomical landmarks across subjects, critical for accurate inter-subject comparisons. Modulation adjusts voxel values to account for local volume changes during normalization, maintaining the integrity of the original tissue properties within the dataset.

Following preprocessing, distinct feature extraction techniques are applied to fMRI and sMRI data to capture relevant neural characteristics. For fMRI, functional connectivity matrices are created by correlating time series data from ROIs defined by the HO atlas. These matrices elucidate the functional interactions and synchrony between various brain regions, pivotal for deciphering the neural basis of ASD. For sMRI, voxel-based morphometry quantifies the density and volume of brain tissues, facilitating the identification of structural differences between ASD participants and controls. This approach is vital for exploring the morphological facets of ASD.

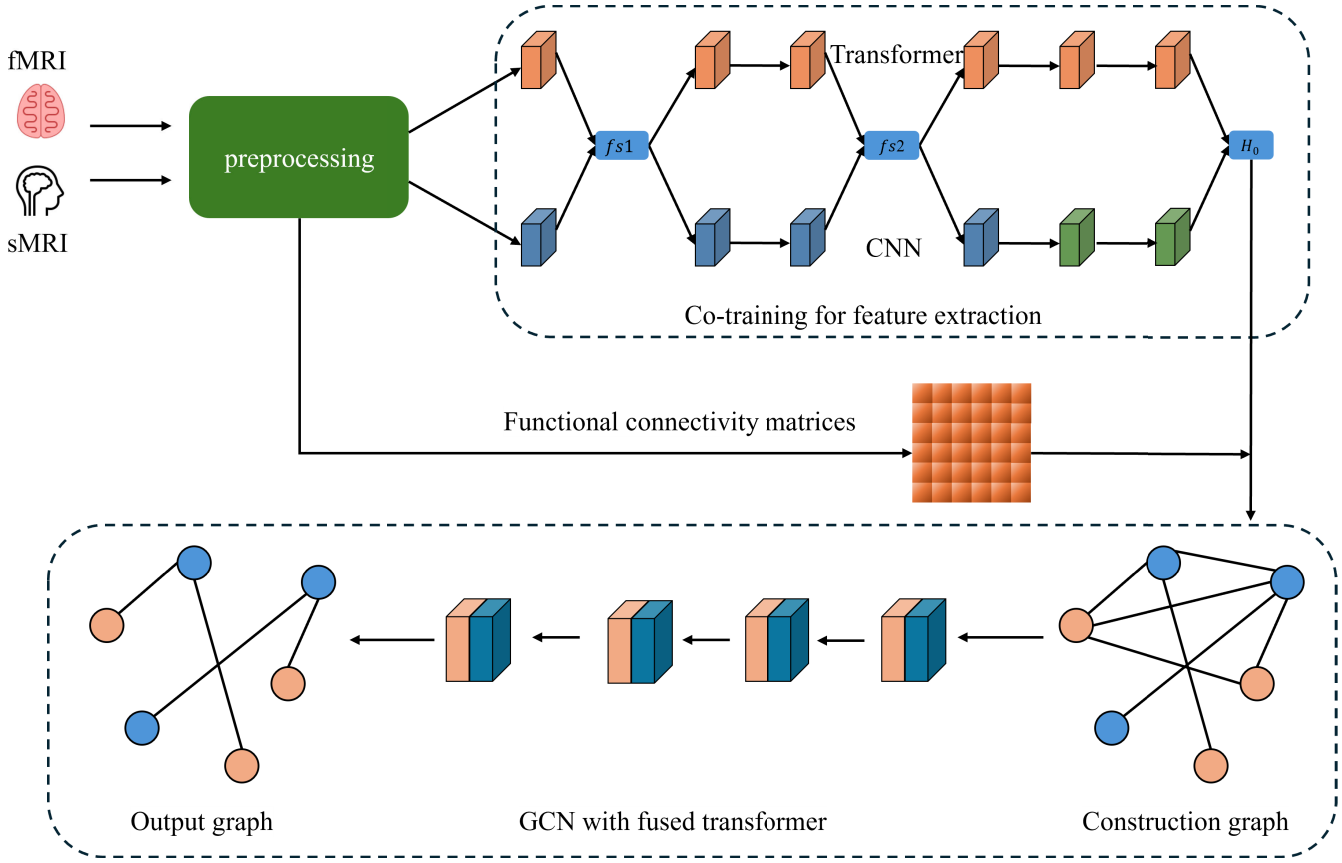


FIGURE 1. Overview of the proposed multimodal ASD classification framework, integrating preprocessing, Transformer-based fMRI feature extraction, CNN-based sMRI feature extraction, and GCN fusion.

In this study, the functional connectivity matrices for fMRI data are computed using Pearson's correlation coefficient to assess the degree of linear relationship between the time series of different brain regions. These ROIs are predefined by the HO brain atlas, facilitating an organized and consistent approach to analyzing brain connectivity.

The Pearson correlation coefficient [22] between two time series X and Y, representing different ROIs, is calculated using the following:

$$r_{XY} = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2}\sqrt{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}} \quad (1)$$

where X_i and Y_i are the values of the time series at time point i. $\bar{X}$ and $\bar{Y}$ are the means of time series X and Y. n is the number of ROIs. This correlation yields a connectivity matrix where each element r_{ij} represents the correlation coefficient between the time series of ROI_i and ROI_j.

The resulting functional connectivity matrix is symmetric (i.e. $r_{ij} = r_{ji}$), with diagonal elements r_{ii} representing the correlation of each ROI with itself, which is always 1. To simplify the analysis and reduce data redundancy. We remove diagonal values. The diagonal elements of the matrix are removed since they do not provide information about inter-regional connectivity. Only the upper triangular portion of the matrix is retained. This part of the matrix captures all unique pairwise correlations, as the matrix's symmetry renders the lower triangular elements redundant. By processing the connectivity matrix in this manner—removing the diagonal and extracting the upper triangular part—we efficiently condense the matrix to include only unique, non-redundant functional connectivity data. This streamlined data representation is essential for subsequent analyses, such as machine learning models that might be employed to distinguish between ASD and control groups based on patterns of brain connectivity. This approach not only ensures that the data fed into analytical models is concise and relevant but also helps in maintaining computational efficiency and clarity in interpreting the models' outputs.

This rigorous preprocessing and feature extraction framework ensure the extraction of high-quality, reliable data from both imaging modalities. Such meticulous data preparation is foundational for the advanced machine learning analyses that follow, aiming to uncover complex neural signatures associated with ASD and enhance our understanding of neurodevelopmental disorders.

B. TRANSFORMER FOR fMRI FEATURE EXTRACTION

To effectively extract and interpret the dynamic interactions within fMRI data, we employ a transformer model [23], specifically designed to handle sequential data and its

inherent complexities. The model begins by converting each time series from the ROIs defined by the HO brain atlas into high-dimensional vectors through an embedding process. This transformation is crucial as it enables the Transformer to process the raw fMRI signals and detect nuanced patterns in the neural activity.

To account for the sequential nature of the data, positional encodings are added to the embeddings. These encodings provide the model with temporal context, an essential component since the Transformer architecture does not inherently recognize the order of input data. The incorporation of positional information allows the model to differentiate between the same neural signals occurring at different times, a key factor in understanding temporal dynamics in brain activity.

The core of the Transformer consists of six encoder layers, each designed to process the data in multiple dimensions. At the heart of each encoder is a multi-head self-attention mechanism that allows the model to focus on different segments of the neural sequences. This mechanism computes attention scores by generating a set of query (Q), key (K), and value (V) vectors from the input vectors. The attention function applies a softmax to the scaled dot products of the queries and keys, providing a set of weights for the values, which represent the importance assigned to each data point:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \qquad (2)$$

where d_k is the dimension of the key vectors. This process enables each position in the encoder to attend to all positions in the previous layer, thus facilitating the integration of information across the entire sequence.

Following the attention mechanism, each position passes through a pointwise feedforward network, which applies two linear transformations separated by a ReLU activation function [24]. These networks are applied independently to each position, allowing for parallel processing that enhances the model's efficiency. Each layer concludes with layer normalization and dropout, strategies employed to stabilize the learning process and prevent overfitting, which are crucial given the complexity and variability of fMRI data.

The final output from the Transformer's encoders is then passed through a linear layer to consolidate the learned features into a compact, fixed-size vector. This vector captures the essential aspects of the functional connectivity and temporal dynamics observed in the fMRI scans, preparing the data for subsequent analytical tasks. This sophisticated feature extraction approach using a transformer model allows for a nuanced understanding of the brain's functional architecture and its variations in conditions like ASD.

C. CNN FOR sMRI FEATURE EXTRACTION

To effectively analyze structural MRI data for indications of structural abnormalities associated with ASD, we have developed a tailored CNN that meticulously captures spatial features reflective of brain structure. The design of this CNN is specifically adapted to handle the complex, three-dimensional nature of sMRI scans, ensuring comprehensive processing and analysis of the brain's anatomical features.

The CNN initiates with an input layer that receives preprocessed sMRI scans. As the scans progress through the network, they encounter four convolutional layers, each equipped with a set of learnable filters. These filters are adept at extracting a variety of spatial features such as edges, textures, and other significant anatomical patterns, which are crucial for identifying potential neural discrepancies linked to ASD. Following each convolutional operation, a ReLU activation function is applied. This function introduces necessary non-linearity into the model, allowing it to learn and model complex patterns that linear models cannot capture, which is essential for effectively representing the intricate structures within the brain.

After each convolutional layer [25], the network applies max pooling to reduce the spatial dimensions of the resulting feature maps. This reduction not only decreases the computational load but also emphasizes the most prominent features, making the model's detections invariant to minor shifts and distortions in the input data. To further enhance model stability and efficiency, batch normalization is employed following each pooling operation. This process normalizes the outputs of the network by adjusting and scaling the activations, thus facilitating faster and more stable training by reducing internal covariate shift.

Towards the end of the CNN, the spatial features extracted by the convolutional and pooling layers are consolidated into a high-level representation through fully connected layers. Specifically, after the final convolutional and pooling operations, the feature maps are flattened into a one-dimensional vector. This vector is then processed through two fully connected layers, which integrate localized information from earlier in the network, synthesizing it into a comprehensive feature representation of the brain's structural data. The first fully connected layer is designed to reduce the dimensionality of the features extracted at $s4$. High-dimensional feature representations often contain redundant information, which can hinder efficient learning. By applying a fully connected layer, we reduce the feature space's dimensionality while retaining the most relevant information. This step is critical for improving computational efficiency and ensuring that the model focuses on the most important features. The second fully connected layer applies a non-linear transformation to the reduced feature set. This layer does not serve as an activation function layer alone, but as a full transformation that combines weights and activation functions to model complex relationships between the features. By performing a non-linear transformation, the second fully connected layer enables the model to learn more intricate patterns that are necessary for accurate prediction or classification.

The final output of the CNN is a feature vector that encapsulates the critical structural characteristics identified in the sMRI scan. This feature vector is not directly used for classification at this stage but rather serves as a compact,

high-level representation of the structural data. It is designed to be combined with the feature vector extracted from the fMRI data by the Transformer model. The combined feature vectors from both the CNN and the Transformer model are then used as inputs for subsequent GCNs, which further analyze the relationships and patterns within the data to classify ASD and control subjects.

Through the strategic layering and configuration of its architecture, the CNN is finely tuned to extract the most salient features from sMRI data, providing valuable insights into the morphological characteristics that may be indicative of ASD. This careful design ensures that the structural complexities of the brain are accurately represented.

D. CO-TRAINING FOR TRANSFORMER AND CNN

In this study, we employ a co-training approach to leverage the complementary information provided by fMRI and sMRI data. This method integrates features from both modalities through a multi-stage fusion strategy, enhancing the overall learning performance. Co-training is particularly advantageous in scenarios where each modality offers unique insights, as it allows for the iterative refinement of feature representations, ultimately leading to a more robust and accurate model. Our implementation uses a Transformer to process fMRI data and a CNN for sMRI data, capitalizing on their respective strengths in handling temporal and spatial information. The model of Co-training is shown in Figure 2.

The process begins with the initial feature extraction from both fMRI and sMRI data. The fMRI data, processed through a six-layer Transformer, yields an initial feature representation denoted as $f1$ after the first Transformer layer. Concurrently, the sMRI data passes through the first convolutional layer of the CNN, producing an initial feature representation $s1$. These two initial representations, $f1$ and $s1$, are then fused to form a combined feature representation $fs1$. This fusion involves concatenating the feature vectors, which preserves the distinct information from each modality while creating a richer, integrated representation.

Following the initial fusion, the combined feature $fs1$ is utilized for further feature extraction. For the fMRI pathway, $fs1$ serves as the input for the second Transformer layer, resulting in the feature $f2$. This process is repeated with $f2$ passing through the third Transformer layer to generate $f3$. In the parallel sMRI pathway, $fs1$ is input into the second convolutional layer of the CNN, leading to the feature $s2$, which is subsequently refined through the third convolutional layer, producing $s3$. At this stage, a second fusion occurs between $f3$ and $s3$, creating a more refined combined feature $fs2$. This second fusion step enhances the representation by integrating the progressively more sophisticated features extracted from both modalities.

The final stages of feature extraction involve further processing of $fs2$ in both the Transformer and CNN architectures. For the Transformer, $fs2$ passes through the fourth and fifth layers, yielding $f4$ and $f5$, respectively. The ultimate fMRI feature representation, F, is obtained after the sixth Transformer layer. Meanwhile, in the CNN pathway, $fs2$ is processed through the fourth convolutional layer, resulting in $s4$. This is followed by two fully connected layers, which transform $s4$ into $s5$ and then into the final sMRI feature representation, S. The fusion feature vector H_0, which combines the final fMRI feature F and the final sMRI feature S, encapsulates the integrated multimodal information and serves as the input for the subsequent GCN.

The co-training and multi-stage fusion strategy adopted in this study not only enhances the predictive performance of the model but also provides a comprehensive framework for understanding the neural underpinnings of ASD. By iteratively refining feature representations and integrating them at various stages, this approach ensures that the final feature vector retains the most informative aspects of both functional and structural brain data. This integrated feature set enables the GCN to effectively classify ASD and control subjects, offering valuable insights into the complex neural signatures associated with the disorder. This methodology not only enhances our understanding of ASD but also sets a precedent for future studies employing multimodal neuroimaging data [26].

E. GCN WITH FUSED TRANSFORMER

In this section, we present a detailed description of the methodology employed to utilize high-dimensional features extracted by the co-training of CNN and Transformer models as inputs for a GCN. The proposed model leverages a four-layer GCN architecture, each layer followed by a Transformer-based self-attention mechanism. The architecture is designed to efficiently capture local and global dependencies in graph-structured data, and the augmented model uses neuroimaging data to classify ASDs.

After co-training the CNN on sMRI data and the Transformer on fMRI data, the high-dimensional feature vectors obtained from both modalities are fused to form a comprehensive representation of the brain's structural and functional characteristics. This fused feature vector, denoted as H_0, serves as the input to the GCN.

The core of the proposed model is a four-layer GCN, where each GCN layer is designed to capture local structural information by performing convolutional operations on the graph-structured data. The input to the l-th GCN layer [27], [28] is the feature matrix H^l from the previous layer. For the first layer, H^0 is the fused feature vector H_0. The graph convolution operation at layer l is defined as:

$$H^{(l+1)} = \sigma(\tilde{A}H^{(l)}W^{(l)}) \tag{3}$$

where $\tilde{A} = D^{-\frac{1}{2}}AD^{-\frac{1}{2}}$ is the normalized adjacency matrix, with A representing the adjacency matrix of the graph and D being the degree matrix. $W^{(l)}$ is the trainable weight matrix at layer l. σ denotes the ReLU activation function.

To capture long-range dependencies and enhance the global context understanding, a Transformer-based self-attention mechanism [23] is applied after each GCN layer.

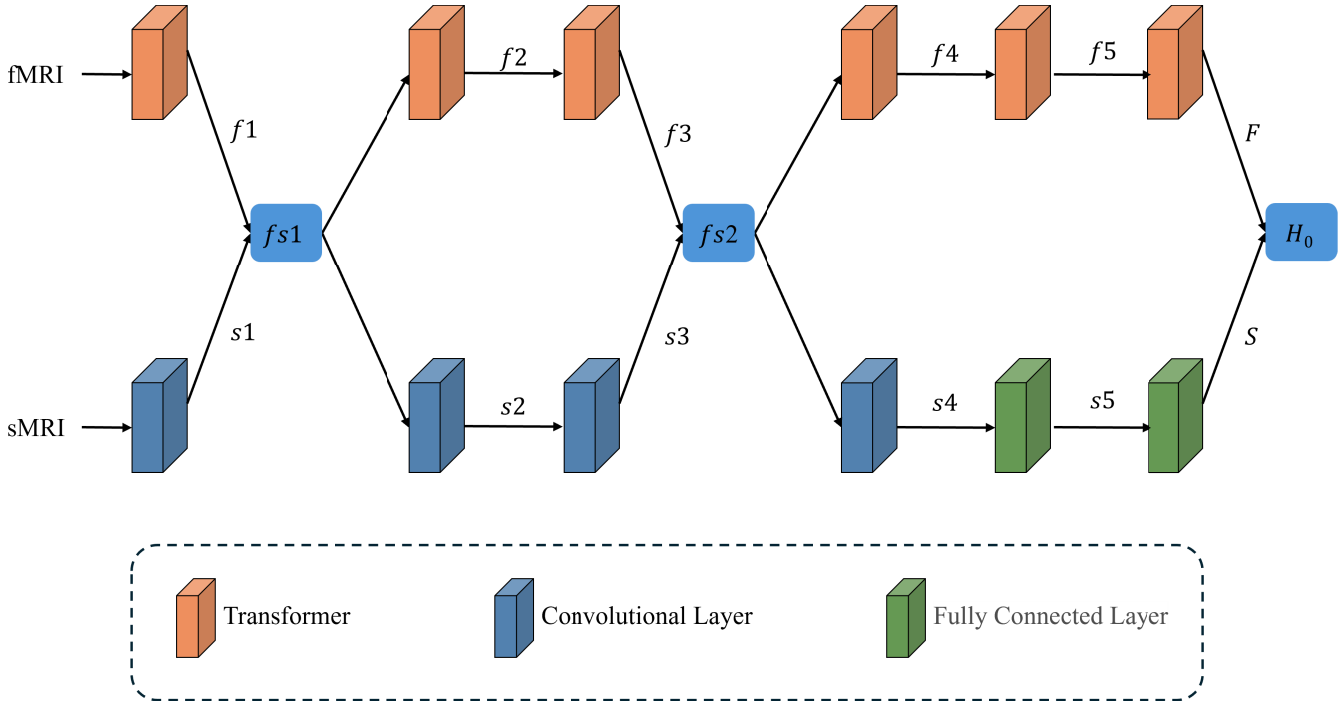


FIGURE 2. Co-training structure combining fMRI features (extracted via Transformers) and sMRI features (extracted via CNNs), with iterative feature fusion steps.

The self-attention mechanism processes the output $H^{(l+1)}$ from the GCN layer:

$$Q^{(l)} = H^{(l+1)}W_Q^{(l)} \tag{4}$$

$$K^{(l)} = H^{(l+1)}W_K^{(l)} \tag{5}$$

$$V^{(l)} = H^{(l+1)}W_V^{(l)} \tag{6}$$

where $W_Q^{(l)}, W_K^{(l)}, W_V^{(l)}$ are trainable weight matrices for the query, key, and value projections, respectively. They are learned weight matrices that correspond to the query, key, and value, respectively. These weight matrices are initialized randomly at the beginning of the training process and are learned during the model's training phase. They are optimized using backpropagation, where the gradients of the loss function with respect to these matrices are computed and updated iteratively through gradient-based optimization algorithms such as Adam. The attention scores are computed as (2). Combined with the features here, it can be expressed as:

$$\text{Attention}^l(Q, K, V) = \text{softmax}\left(\frac{Q^{(l)}(K^{(l)})^T}{\sqrt{d_k}}\right)V^{(l)} \tag{7}$$

where d_k is the dimension of the key vectors. The output of the self-attention mechanism is then added back to the input of the GCN layer (residual connection) and passed through a feedforward network [29]:

$$H_{attention}^{(l+1)} = FFN(Attention^l(Q, K, V) + H^{(l+1)}) \tag{8}$$

where FFN represents a feedforward neural network. This process is repeated for four layers, with each layer's output being fed into the subsequent GCN layer followed by the Transformer self-attention mechanism. The model of GCN fusion transformer is shown in Figure 3.

As can be seen from Figure 3, the output graph has fewer edges than the input graph, which reflects the application of the Edgedrop strategy [13]. This technique, where a certain proportion of edges are randomly deleted during training, helps increase the diversity of the graph at each layer and reduces the risk of overfitting. By introducing this randomness, the model is encouraged to learn more robust features, preventing it from memorizing specific relationships in the data that may be unique to the training set.

In the construction of graph convolutional networks, the edges of the graph are defined by the functional connectivity matrix, which is calculated based on the Pearson correlation coefficients between the time series of different brain regions in the fMRI data. This is shown in (1). The proposed four-layer GCN with integrated Transformer self-attention mechanisms allows the model to effectively capture both local and global patterns within the graph-structured data. This architecture is particularly well-suited for complex tasks like ASD classification, where understanding both fine-grained and holistic brain connectivity patterns is crucial. The combination of GCN's local feature learning and Transformer's global attention ensures that the model has a comprehensive view of the underlying neuroimaging data, leading to more accurate and robust predictions.

IV. EXPERIMENTS AND RESULTS ANALYSIS

A. EXPERIMENTAL SETTINGS

In our experimental setup, we employed nested ten-fold cross-validation on the ABIDE-I dataset, which offers a

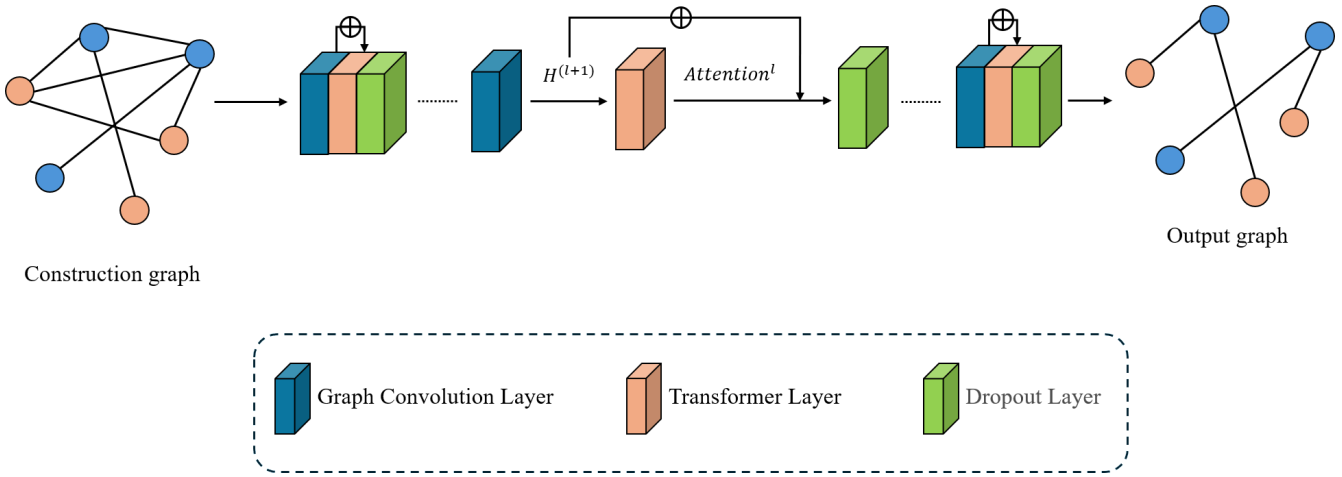


FIGURE 3. GCN fusion transformer architecture, integrating GCN layers and transformer-based self-attention to model both local and global brain connectivity.

more robust evaluation compared to standard ten-fold cross-validation [30] by incorporating an additional layer of validation to prevent overfitting and ensure the generalizability of the model. Specifically, the outer loop of the cross-validation is used to assess model performance, while the inner loop optimizes the hyperparameters, providing a more reliable estimate of the model's true performance. In the experimental settings, we also employed leave-one-out cross-validation (LOO-CV) [31] on a subset of the ABIDE-II dataset. LOO-CV is a rigorous method where each sample is used as a validation set while the remaining samples form the training set, ensuring that every observation is tested. This approach is particularly useful for small datasets, as it maximizes the use of available data and provides a reliable estimate of model performance, making it ideal for evaluating the robustness of our model on this specific dataset.

The parameter settings for each module are shown in Table 1. The input feature vector $H_(0)$ to the GCN was constructed with 3000 dimensions, representing the high-dimensional features extracted and fused from the fMRI and sMRI data. The GCN-based model was trained over 300 epochs using the Adam optimizer, which is well-suited for large-scale neural network training due to its adaptive learning rate capabilities. The learning rate was set to 0.01, and a dropout rate of 0.2 was applied to prevent overfitting [32] by randomly dropping units during training. All models were implemented using the open-source machine learning library PyTorch, ensuring flexibility and reproducibility in the experimental process. The training and evaluations were conducted on a GeForce GTX 4060 GPU, which provided the necessary computational power for handling the high-dimensional data and complex model architecture.

For performance evaluation [33], [34], we employed several key metrics: Accuracy, Precision, Recall, and the Area Under the Curve (AUC) of the Receiver Operating Characteristic (ROC) curve. These metrics provide a comprehensive

TABLE 1. Hyperparameter configurations for the Transformer, CNN, and GCN Module used in the proposed model.

Module	Parameter	Value
Transformer	Number of layers	6
	Number of attention heads	8
	Hidden size	512
	Activation function	ReLU
	Dropout rate	0.2
CNN	Number of convolutional layers	4
	Filter size	3×3
	Activation function	ReLU
	Dropout rate	0.2
GCN	Number of layers	4
	Number of hidden units	128
	Activation function	ReLU
	Dropout rate	0.2

assessment of the model's classification performance. Their mathematical definitions are shown below:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{9}$$

$$Precision = \frac{TP}{TP + FP} \tag{10}$$

$$Recall = \frac{TP}{TP + FN} \tag{11}$$

where TP refers to the correct classification of positive cases. TN refers to the correct classification of negative cases. FP refers to the incorrect prediction of a positive case. FN refers to the incorrect prediction of a negative case. Accuracy measures the proportion of correct predictions across all samples. Precision indicates the proportion of true positive classifications among all samples predicted as positive, reflecting the model's ability to avoid false positives. Recall, also known as sensitivity, measures the proportion of actual positives that were correctly identified by the model, highlighting its capability to detect all relevant instances. Lastly, the AUC represents the area under the ROC curve, providing a single scalar value that summarizes the model's

ability to discriminate between positive and negative classes. A higher AUC indicates better model performance, with a value closer to 1.0 signifying a highly effective classification method.

B. ABLATION STUDY

The ablation study was meticulously designed to evaluate the contribution of different components of our proposed model by systematically altering the model's architecture and observing the resulting changes in performance metrics. We conducted four sets of experiments, each progressively incorporating more advanced features of our model. The first configuration serves as our baseline, while subsequent configurations introduce additional components to assess their impact.

In the first configuration, the fMRI and sMRI data, after undergoing preprocessing, including Recursive Feature Elimination (RFE) [35] and Principal Component Analysis (PCA), are concatenated to form the input to the GCN. The model does not integrate any Transformer components in this setup. This experiment establishes the baseline performance, providing a reference point for evaluating the impact of adding Transformer-based components in subsequent experiments. In the second setup, the model remains similar to the baseline GCN, but it includes Transformer components after the GCN layers. Here, the preprocessing remains unchanged, with early-stage feature extraction techniques like RFE or PCA applied beyond the standard preprocessing. The purpose of this experiment is to highlight the significance of incorporating the Transformer for enhancing the model's ability to capture global context in the classification process. The third configuration involves the use of a Transformer specifically for feature extraction from the fMRI data. However, unlike the full proposed model, this setup does not utilize Co-training between fMRI and sMRI data. The GCN is then applied to the concatenated features, followed by the Transformer integration similar to the previous setup. This experiment is designed to demonstrate the importance of the Transformer during the feature extraction phase, assessing its direct impact on the quality of features provided to the GCN. The fourth settings is the complete model proposed in this study, incorporating all components: the preprocessing for fMRI and sMRI, the co-training of CNN on sMRI data and transformer on fMRI data, followed by the integration of these features into the GCN, and subsequent Transformer layers. The full model allows us to assess the cumulative benefits of all the advanced techniques and the importance of co-training in enhancing classification performance. The experimental results for each group are shown in Table 2.

The ablation study provides both qualitative and quantitative insights into the contribution of each component in our proposed model architecture. By systematically varying the model configurations, we can assess the individual and combined effects of different elements, including the GCN layers, Transformer integration, and the co-training process.

C. PERFORMANCE EVALUATION

To ensure a fair comparison, we selected baseline models with similar objectives: HOFC [36], ASD-DiagNet [37], and GCN [38]. Furthermore, to demonstrate the superiority of our proposed method, we benchmarked it against several state-of-the-art (SOTA) approaches, including MVS-GCN [39], the Spatial-Temporal Transformer [40], and HMST [41].

HOFC focuses on improving the sensitivity of functional connectivity analysis by incorporating higher-order information. This method compares traditional functional connectivity with HOFC, where HOFC is demonstrated to have a higher sensitivity in detecting group differences, especially at stringent statistical thresholds. HOFC builds upon the pairwise FC by calculating the higher-order correlations between these pairwise connections, effectively capturing more complex relationships between brain regions.

ASD-DiagNet utilizes a combination of feature extraction techniques and a hybrid model that employs CNNs for feature learning and SVMs for classification. The key innovation in ASD-DiagNet lies in its ability to balance the dataset using data augmentation techniques, which are critical in achieving more accurate and generalized classification results across different sites and age groups.

GCN employs the relationships between different brain regions based on functional connectivity data. The approach constructs a graph where each node represents a brain region, and the edges represent the functional connectivity between these regions. The GCN is used to learn features that capture the global structure of brain connectivity, which are then fed into a classifier to distinguish between different neurological conditions.

MVS-GCN integrates multi-view learning with spectral graph convolutional networks. This method uses multiple views of the data, each representing different aspects of the brain's functional and structural connectivity, and applies spectral GCNs to each view. The outputs from these GCNs are then fused to form a comprehensive representation of the brain's connectivity patterns, which is used for classification.

Spatial-Temporal Transformer utilizes a transformer-based architecture to process both spatial and temporal information in fMRI data. The model employs multi-headed attention mechanisms to separately capture spatial and temporal dependencies, which are then combined to generate a unified representation for classification. The use of transformers allows this method to model complex interactions within the data more effectively than traditional convolutional approaches.

HMST This approach applies a hierarchical strategy to process fMRI data at different scales, capturing both fine-grained and coarse-grained patterns in the data. The multi-scale aspect allows the model to adaptively focus on different levels of information, while the hierarchical structure helps in capturing long-range dependencies more effectively. This method stands out due to its ability to model both local and global interactions within the brain's functional connectivity data.

TABLE 2. Results of the ablation study, examining the contribution of different model components to ASD classification performance.

Method	Accuracy (%)	Precision (%)	Recall (%)	AUC
GCN	71.85	70.08	75.32	0.75
GCN+Transformer	75.62	74.21	77.89	0.78
GCN+Dual Transformer	77.93	76.85	79.51	0.81
Ours	79.47	78.97	82.11	0.85

TABLE 3. Comparison of ASD performance on the ABIDE-I dataset against various baseline methods.

Method	Accuracy (%)	Precision (%)	Recall (%)	AUC
HOFC	70.12	68.75	71.50	0.72
ASD-DiagNet	72.30	71.00	73.40	0.74
GCN	71.85	70.08	75.32	0.75
MVS-GCN	72.08	68.50	73.20	0.72
Spatial-Temporal Transformer	77.10	76.00	78.00	0.80
HMST	78.50	77.80	79.30	0.82
Ours	79.47	78.97	82.11	0.85

Replication of different models based on codes and papers provided by the researchers, and nested ten-fold cross-validation in the same setup. The experimental results are shown in Table 3.

The performance evaluation across seven different methods provides a comprehensive understanding of how each approach contributes to the classification of ASD using fMRI and sMRI data. The results highlight the strengths and limitations of both baseline and SOTA methods, as well as the superiority of the proposed method in leveraging a combined GCN and Transformer-based architecture with co-training.

D. LEAVE-ONE-OUT CROSS-VALIDATION

In this section, we applied LOO-CV to evaluate our proposed method on a subset of the ABIDE-II dataset, which includes 92 ASD subjects and 103 neurotypical controls. LOO-CV is a stringent evaluation method where each sample is used as a validation set once, while the remaining samples form the training set. This process ensures that the model is tested on every possible division of the data, providing a highly reliable measure of its generalization ability.

The performance of our method was compared against two other approaches. The first is STCAL [42], which, as described earlier, employs a co-attention learning strategy to effectively capture spatial and temporal dependencies in the fMRI data. The second is the Spatial-Temporal Transformer [40], which also models spatial and temporal information but lacks the advanced co-attention mechanisms of STCAL. Finally, our proposed method, while slightly lower in performance compared to its results on the ABIDE-I dataset due to the use of LOO-CV, still demonstrates its robustness and effectiveness. The results of the comparison experiment are shown in Figure 4.

The LOO-CV results on the ABIDE-II dataset provide valuable insights into the robustness and effectiveness of our proposed method compared to other state-of-the-art approaches.

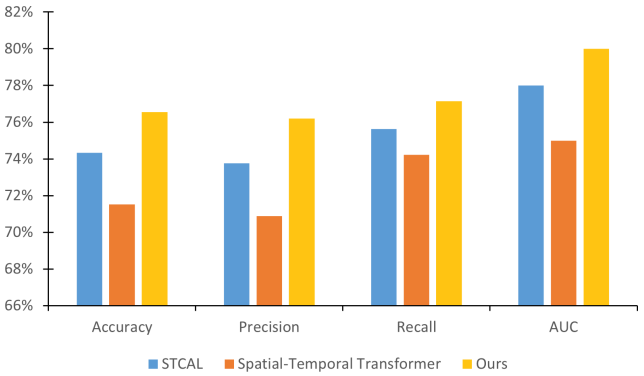


FIGURE 4. Comparison of performance on the ABIDE-II dataset, including Accuracy, Precision, Recall, and AUC against baseline methods.

V. DISCUSSION

As shown in Table 2, starting with the GCN (Baseline) configuration, we observe that while the model can achieve a reasonable level of performance (Accuracy 71.85%, AUC 0.75), it primarily captures local structural information due to its reliance on features processed through RFE and PCA. This configuration lacks the ability to capture the more complex, global dependencies present in the data, which limits its effectiveness in accurately classifying the input data. The relatively lower recall of 75.32% indicates that while the model can identify a fair number of true positive cases, it may struggle with capturing the full spectrum of relevant patterns, leading to missed detections.

In the GCN + Transformer setup, the integration of Transformer components after the GCN layers introduces the ability to model long-range dependencies and capture global context within the graph structure. The result is a noticeable improvement in performance, with the model achieving an accuracy of 75.62% and an AUC of 0.78. The precision increases to 74.21%, reflecting a reduction in false positive predictions, which suggests that the model is better at distinguishing between subtle differences in the data. The improvement in recall (77.89%) also indicates

that the Transformer is effective in enhancing the model's sensitivity to relevant patterns, allowing it to identify a greater proportion of true positive cases.

The GCN + Dual Transformer configuration further emphasizes the importance of the Transformer in the feature extraction process, specifically for handling fMRI data. By using the Transformer to process fMRI data before it is input into the GCN, we see a more substantial improvement in performance metrics: accuracy reaches 77.93% and AUC 0.81. The precision of 76.85% and recall of 79.51% indicate that this configuration benefits from the Transformer's ability to capture complex temporal dynamics, leading to a more refined and effective feature representation. The dual Transformer approach also suggests that even without co-training, the model's ability to integrate information from different brain regions is significantly enhanced, contributing to better classification performance.

Finally, the Ours configuration, which incorporates co-training, achieves the highest performance across all metrics, with an accuracy of 79.47%, precision of 78.97%, recall of 82.11%, and AUC of 0.85. This comprehensive setup demonstrates the cumulative benefits of all components working together. The co-training process, in particular, plays a crucial role in refining the features extracted from both fMRI and sMRI data, leading to better generalization and robustness in the model's predictions. The high recall of 82.11% in this configuration highlights the model's superior sensitivity, indicating that it is adept at identifying true positive cases while minimizing the risk of false negatives. The increased AUC value further underscores the model's overall effectiveness in distinguishing between classes, suggesting that the combination of GCN layers, Transformer self-attention, and co-training enables the model to achieve a well-balanced performance across all evaluation metrics.

In summary, the ablation study quantitatively shows that each component of our model contributes to incremental improvements in performance, with the most significant gains observed when all components are integrated. Qualitatively, the study reinforces the importance of using both local and global context modeling (through GCN and Transformer, respectively) and the role of co-training in leveraging complementary data sources (fMRI and sMRI). These findings validate the superiority of the proposed architecture in addressing the complex classification tasks inherent in neuroimaging data, particularly in the context of ASD classification.

As shown in Table 3, starting with the baseline methods, HOFC shows the lowest performance with an accuracy of 70.12% and an AUC of 0.72. While HOFC offers a novel approach by incorporating higher-order information into functional connectivity analysis, it struggles to capture the more complex patterns in the data that are necessary for accurate ASD classification. This is reflected in the lower precision and recall values, indicating challenges in both identifying true positives and avoiding false positives.

ASD-DiagNet improves upon HOFC with an accuracy of 72.30% and an AUC of 0.74. The hybrid learning approach, combining CNNs and SVMs, provides a balanced methodology for feature learning and classification. However, its reliance on traditional machine learning techniques such as SVMs limits its ability to fully exploit the rich, high-dimensional data available in neuroimaging, leading to moderate improvements over HOFC.

GCN further advances the performance, achieving an accuracy of 71.85% and an AUC of 0.75. By modeling the relationships between different brain regions through functional connectivity graphs, this method better captures the global structure of the brain's connectivity. However, the absence of a mechanism to capture temporal dependencies limits its overall performance, particularly in distinguishing subtle patterns in the data, as evidenced by its performance metrics.

Transitioning to the SOTA methods, MVS-GCN demonstrates significant improvements with an accuracy of 72.08% and an AUC of 0.72. The integration of multi-view learning with spectral GCNs allows this method to combine different types of data, thereby capturing more nuanced patterns associated with neurological disorders.

Spatial-Temporal Transformer further enhances performance, achieving an accuracy of 77.10% and an AUC of 0.80. By utilizing a transformer-based architecture to process both spatial and temporal information, this method excels in modeling the complex interactions within the data. The multi-headed attention mechanism captures detailed dependencies, leading to better overall classification results.

HMST pushes the performance even higher, with an accuracy of 78.50% and an AUC of 0.82. The hierarchical multi-scale approach enables the model to adaptively focus on different levels of information, capturing both local and global interactions more effectively. This results in the highest performance among the baseline and other SOTA methods, underscoring the importance of multi-scale analysis in neuroimaging.

Finally, the proposed method, Ours, achieves the highest performance across all metrics, with an accuracy of 79.47%, precision of 78.97%, recall of 82.11%, and AUC of 0.85. These results demonstrate the effectiveness of combining GCN and Transformer-based architectures with co-training. The quantitative improvements over both baseline and SOTA methods are significant, particularly in recall and AUC, indicating that the proposed method not only captures the complex patterns in the data more effectively but also generalizes better across different samples.

From a qualitative perspective, the discussion highlights the strengths of each method. HOFC brings an innovative approach to functional connectivity analysis, yet its inability to handle the high-dimensional and complex nature of neuroimaging data limits its utility in clinical applications. ASD-DiagNet benefits from a hybrid approach, but its reliance on traditional SVMs constrains its capacity to fully exploit deep learning's potential in feature extraction. GCN

introduces a significant improvement by leveraging graph structures, providing a more holistic understanding of brain connectivity. However, its lack of temporal modeling is a key shortcoming. On the other hand, MVS-GCN successfully integrates multi-view data, which adds a new dimension to the analysis, while Spatial-Temporal Transformer and HMST build upon this by incorporating advanced transformer architectures that excel at capturing both spatial and temporal dynamics in the data.

The superior performance of Ours underscores the critical role of co-training in effectively integrating multimodal data (fMRI and sMRI) and the advantages of combining GCN's structural learning with Transformer's powerful context modeling. The co-training mechanism is particularly crucial in balancing and refining the features extracted from both modalities, leading to a model that is not only more accurate but also more robust in its predictions.

Overall, the discussion illustrates that while each method has its merits, the proposed method's ability to integrate the strengths of GCN and Transformer architectures with co-training sets it apart, leading to superior performance in ASD classification tasks. The combination of advanced techniques in the proposed method allows for a more comprehensive and nuanced analysis of neuroimaging data, ultimately contributing to better clinical outcomes.

Further observation of LOO-CV results shows that, as shown in Figure 4, the quantitative results show that our method outperformed both the STCAL and Spatial-Temporal Transformer approaches across all evaluation metrics. Specifically, our method achieved an accuracy of 76.55%, a precision of 76.20%, a recall of 77.14%, and an AUC of 0.80. These results, while slightly lower than those obtained on the ABIDE-I dataset, reflect the rigorous nature of LOO-CV, which is known for providing a thorough and challenging assessment of model performance.

STCAL achieved an accuracy of 74.33%, a precision of 73.77%, a recall of 75.62%, and an AUC of 0.78. Although STCAL benefits from its co-attention learning strategy, which effectively captures both spatial and temporal dependencies, it falls short of our method's performance. This suggests that while STCAL is robust in identifying key regions of interest and time frames, it may not fully leverage the complex interactions present in the data.

Spatial-Temporal Transformer, on the other hand, exhibited lower performance with an accuracy of 71.53%, a precision of 70.88%, a recall of 74.21%, and an AUC of 0.75. This method, while effective in modeling spatial and temporal information, lacks the co-attention mechanisms present in STCAL and our approach, which likely contributes to its lower performance. The reduced precision and recall indicate that this method may struggle more with correctly identifying ASD cases and avoiding false positives.

From a qualitative standpoint, the superior performance of our method can be attributed to several key factors. First, the integration of GCN and Transformer architectures enables our model to capture both local and global dependencies in the data. The GCN effectively models the structural relationships between brain regions, while the Transformer captures complex temporal dynamics. This combination is particularly powerful in handling the multifaceted nature of neuroimaging data, where both spatial and temporal patterns play crucial roles.

Second, the co-training approach employed in our method is essential in refining the features extracted from fMRI and sMRI data. By allowing the model to learn from both modalities simultaneously, co-training enhances the model's ability to generalize across different samples, which is crucial in a leave-one-out cross-validation setting where each sample is tested individually. This likely contributes to the model's higher recall, as it becomes more adept at capturing the diverse presentations of ASD.

In contrast, STCAL's reliance on separate attention mechanisms for spatial and temporal information, while effective, may not fully exploit the interdependencies between these aspects. Similarly, the Spatial-Temporal Transformer's lack of advanced co-attention mechanisms limits its ability to model the data as effectively, resulting in lower overall performance.

Overall, the discussion highlights that while each method has its strengths, our proposed approach stands out due to its comprehensive integration of advanced techniques. The combination of GCN, Transformer, and co-training allows our method to achieve superior performance, particularly in challenging validation scenarios like LOO-CV. This robustness makes it a promising tool for clinical applications where accurate and reliable classification is paramount.

VI. CONCLUSION

In this study, we proposed a novel approach that integrates GCN and dual transformer-based architectures, enhanced with a co-training strategy, for the classification of ASD using fMRI and sMRI data. Through rigorous evaluation, including nested ten-fold cross-validation on the ABIDE-I dataset and leave-one-out cross-validation on the ABIDE-II dataset, our method demonstrated superior performance compared to both baseline and SOTA models. The results highlight the effectiveness of combining GCN's structural learning with the Transformer's ability to capture complex temporal dynamics. Additionally, the co-training approach significantly improved the model's robustness and generalization capabilities. These findings underscore the potential of our method as a powerful tool for neuroimaging-based ASD diagnosis, offering valuable insights for future research and clinical applications.

While our proposed method shows significant improvements in ASD classification, it has some limitations. The model's performance can be influenced by the variability in fMRI and sMRI data, such as differences in scanning protocols and demographic diversity, which may affect generalizability. Additionally, the complexity of integrating GCN and Transformer architectures with co-training could pose challenges in computational efficiency, especially for

larger datasets. Future research should focus on improving data normalization techniques, optimizing the model for better computational efficiency, and exploring the integration of additional neuroimaging modalities or genomic data. Additionally, longitudinal studies could provide deeper insights into the progression of ASD, enhancing the model's clinical utility.

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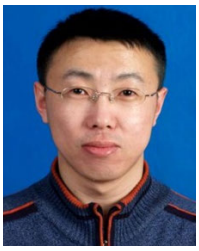
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TIANMING SONG received the B.S. degree in medical image processing from the School of Electrical and Information Engineering, Hei Long Jiang College of Engineering, in 2012, the master's degree from the School of Automation, Harbin University of Science and Technology, in 2015, and the Ph.D. degree from the School of Measurement and Control Technology and Communication Engineering, Harbin Institute of Technology, in June 2022. He is an Associate Professor with Wuxi Vocational College of Science and Technology. His main research interests include artificial intelligence and medical image processing.



ZHE REN received the bachelor's and master's degrees in control science and engineering and the Ph.D. degree in engineering from Harbin University of Science and Technology, in 2012, 2016, and June 2022, respectively. She is a Lecturer with Wuxi Vocational College of Science and Technology. Her primary research interests include artificial intelligence, semantic segmentation, and medical image processing.



JIAN ZHANG received the master's degree in computer applications and the Ph.D. degree in testing and measuring technology and instruments from Harbin Institute of Technology. He was a Visiting Scholar with the University of Pittsburgh, USA, from 2010 to 2011. Currently, he is a Professor with the School of Internet of Things and Artificial Intelligence, Wuxi Vocational College of Science and Technology. His research interests include medical image processing artificial intelligence technologies, blockchain technologies, and computer application technologies.



YAWEI QU received the bachelor's degree in clinical medicine from Taishan Medical University, in 2009, and the master's degree in gastroenterology from Jinzhou Medical University, in 2013. He is currently pursuing the Ph.D. degree in control science and engineering with Harbin Institute of Technology. He is affiliated with the Department of Gastroenterology, Beijing Mentougou District Hospital, Beijing, China.



YINGYING CUI received the bachelor's degree in clinical medicine from Chengde Medical College, in 2010, and the master's degree in gastroenterology from the PLA Medical College, in 2013. She is currently affiliated with the Department of Gastroenterology, Beijing Mentougou District Hospital, Beijing, China.



ZHENGDA LIANG was born in Qingyuan, Guangdong, China, in 1996. He received the Bachelor of Medicine degree from Jining Medical College, in 2018. His main research interest includes medical artificial intelligence.